The Nonlinear Effects of Macroeconomic Uncertainty on Commodity Price Volatility: A Replication and Extension of Joëts et al. (2017)

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1 Abstract

This project revisits the nonlinear relationship between macroeconomic uncertainty and commodity price volatility by replicating and extending Joëts et al. (2017).This study replicates Joëts et al. (2017) by estimating a threshold VAR model over 1986–2015 using the CBOE S&P 100 Volatility Index (VXO), which covers the full replication sample. To extend the analysis through April 2025, the CBOE Volatility Index (VIX), based on the S&P 500, is employed from 1990 onward due to the discontinuation of VXO after August 2020 and the broader market coverage of VIX. For the 1986–1989 period, only VXO is available and used. Additionally, the Jurado-Ludvigson-Ng (JLN) macroeconomic uncertainty index and the ECB's Composite Indicator of Systemic Stress (CISS) are incorporated over the full 1986–2025 sample to capture broader systemic uncertainty.To evaluate model robustness, results are benchmarked across linear and structural VAR frameworks, varying lag lengths, threshold parameters, and bootstrap confidence intervals. Furthermore, sensitivity analyses contrast volatility-based and predictability-based measures of uncertainty to ensure findings hold under alternative proxies. This approach clarifies the influence of uncertainty shocks on commodity markets in contemporary macro-financial conditions, enhancing the reliability of stress testing, risk management, and policy calibration frameworks.

2 Introduction and Motivation

Understanding how macroeconomic uncertainty propagates through commodity markets is critical for modeling risk in today's structurally shifted macro-financial environment. Joëts et al. (2017) examine this relationship using a threshold VAR (TVAR) model applied to 19 monthly commodity return series across energy, precious metals, agriculture, and industrial sectors. While the full dataset spans 1976 to 2015, the empirical analysis is restricted to the post-1986 period due to data availability for their primary uncertainty proxy—the implied volatility index on the CBOE S&P 100 Volatility Index (VXO). Robustness checks are conducted using macro uncertainty indices from Jurado, Ludvigson, and Ng's dataset.

This study utilizes a comprehensive dataset comprising monthly returns on 19 commodity price series across energy, precious metals, agricultural, and industrial metals markets, thereby capturing a wide spectrum of commodity asset classes sensitive to macroeconomic uncertainty.

Traditional linear VAR models assume symmetric responses across economic states, limiting their ability to capture risk asymmetries during turbulent periods. As a result, they may underestimate downside risk and obscure key transmission channels. In contrast, TVAR models allow for regime shifts based on threshold variables. When combined with generalized impulse response functions (GIRFs), they permit the estimation of state-contingent responses to uncertainty shocks—better reflecting real-world volatility dynamics.

Three key extensions are introduced to the original TVAR framework. First, the sample is extended through April 2025 to incorporate recent macro-financial volatility. Second, regime identification is refined using updated uncertainty proxies: the Jurado-Ludvigson-Ng (JLN) index, the ECB's Composite Indicator of Systemic Stress (CISS), and a hybrid volatility-based measure that uses VXO from 1986 to 1989 and VIX from 1990 onward—ensuring continuous coverage beyond the original study's period. Third, TVAR results are benchmarked against linear and structural VAR models, with robustness tests conducted across lag orders, threshold values, and bootstrap intervals.

The hypothesis is that uncertainty shocks—particularly those identified through macroeconomic and systemic stress proxies—have become more asymmetric and persistent since 2015, amplifying commodity price volatility more strongly in high-uncertainty regimes. This nonlinear propagation is expected to vary across commodity groups and may reveal structural shifts not captured by linear models or volatility-based proxies alone.

3 Methodology

The empirical strategy follows Joëts et al. (2017) and is structured around three modules: replication, extension, and validation.

3.1 Replication

The first step replicates the nonlinear framework of Joëts et al. (2017), who estimate a threshold vector autoregressive (TVAR) model to analyze how uncertainty shocks affect commodity price volatility in different regimes. The TVAR allows for distinct autoregressive dynamics depending on whether an uncertainty measure crosses an estimated threshold value. The model is defined as:

$$\mathbf{y}_t = \begin{cases} \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \epsilon_t, & \text{if } q_{t-d} \leq \theta \\ \\ \mathbf{B}_1\mathbf{y}_{t-1} + \cdots + \mathbf{B}_p\mathbf{y}_{t-p} + \epsilon_t, & \text{if } q_{t-d} > \theta \end{cases}$$

Here, $\mathbf{y}_t$ is a vector of monthly log returns on 19 globally traded commodities, covering the energy, precious metals, agricultural, and industrial metals sectors. This matches the full cross-sectional scope of Joëts et al. (2017) and allows for sector-specific analysis of regime-dependent volatility transmission. The commodities and their data spans are as follows:

- Energy markets: WTI crude oil and Henry Hub natural gas (NYMEX), Oct. 1978 – Apr. 2015
- Precious metals: Gold (COMEX), platinum (LME), silver (COMEX), Feb. 1976 – Apr. 2015
- Agriculture: Cocoa (ICO), coffee (UNCTAD), corn (COMEX), cotton (IMF, CIF Liverpool), lumber (IMF), soybeans (COMEX), sugar (COMEX), wheat (MGEX), Feb. 1980 – Apr. 2015
- Industrial metals: Aluminium (LME), copper, lead, nickel, tin, zinc (IMF/LME), Feb. 1980 – Apr. 2015

All price series are transformed via first-logarithmic differences to compute monthly returns. The dataset is aligned to a common monthly frequency and standardized prior to estimation to ensure comparability across series. Where price series are discontinued or missing post-2015, proxy series or interpolated benchmarks (e.g., futures prices, UNCTAD Data Hub) will be used with proper documentation

The threshold variable q_{t-d} is the lagged value of the VXO, smoothed using a three-period moving average to mitigate endogeneity. The threshold level θ is estimated endogenously via grid search, with a 15% trimming applied to ensure sufficient observations in both regimes. The lag order $p = 3$ is chosen in line with the original study, based on standard information criteria. The delay parameter d is determined empirically and tested for robustness.

Estimation is conducted using the `tsDyn` package in R. GIRFs, simulated via the `tvarGIRF` package, trace the average response to a one-standard-deviation shock conditional on regime. The replication sample spans January 1986 to April 2015, consistent with Joëts et al. (2017).

The baseline threshold variable is the VXO, available from 1986 to 1989. From 1990 onward, it is replaced by the VIX (based on S&P 500 options) to ensure continuity in volatility-based regime identification. Both indices are obtained from the CBOE. These series are standardized and lagged with smoothing to define high-uncertainty regimes.

3.2 Extension

The analysis extends Joëts et al. (2017) in two directions: (1) by expanding the sample and uncertainty proxies, and (2) by introducing methodological enhancements to improve robustness and interpretability.

3.2.1 Sample and Proxy Extension

The baseline sample is extended through April 2025 to capture recent macro-financial volatility, including the COVID-19 shock and the inflation-driven tightening cycle of 2022–2024. Post-2015 data will be sourced from FRED, IMF Data, World Bank Pink Sheets, and ECB's Data Portal, subject to availability. Updated uncertainty proxies are incorporated as threshold variables: - VIX (1990–2025) - JLN (1986–2025) - CISS (1986–2025)

TVARs are re-estimated over the appropriate sub-samples using a grid of lag lengths ($p = 2$ to 6) and delay parameters ($d = 1$ to 3), with thresholds θ determined endogenously.

3.2.2 Methodological Enhancements (Prioritized vs. Exploratory)

To balance ambition and feasibility, planned extensions are divided into core and exploratory tracks:

3.2.2.1 Minimum Viable Enhancements (Planned Core Tasks)

- Estimate and compare threshold regimes using VIX, JLN, and CISS.
- Benchmark TVAR results against linear VAR and SVAR baselines.
- Assess GIRF asymmetries across regimes for each proxy via simulation-based mean-path comparisons. Quantile-based impulse comparisons are considered exploratory

3.2.2.2 Exploratory Extensions (Time-Permitting)

- Estimate TVARX specifications to incorporate global spillovers or exogenous controls.
- Experiment with alternative threshold specifications (e.g., rolling volatility windows).
- Explore quantile-specific GIRFs to assess tail risk amplification.
- Implement fixed-design wild bootstrap methods for GIRF inference under heteroskedasticity.

These extensions will be prioritized based on feasibility within the semester's timeframe.

3.3 Validation

To benchmark the value of nonlinear modeling, we compare TVAR results against both linear VAR and structural VAR (SVAR) baselines. The linear VAR is estimated on the same variable set and lag order. For the SVAR, we use recursive identification via Cholesky decomposition, ordering the uncertainty proxy last to isolate its shock. Impulse responses from the linear and structural models are contrasted with regime-contingent GIRFs to assess whether linear frameworks understate volatility transmission in high-risk environments.

3.4 Implementation and Reproducibility

All empirical work will be implemented in R using a modular codebase. Estimation, simulation, and validation procedures are structured into separate scripts to ensure full transparency. The complete codebase—including data preprocessing routines and diagnostic outputs will be archived in a publicly accessible GitHub repository and fully documented for replication and future research.

The empirical implementation will proceed in stages: the core TVAR replication and VIX-based extension will be prioritized initially. Incorporation of additional proxies (JLN, CISS) and advanced methodological enhancements will depend on feasibility and time constraints.

4 Research Contribution

This study extends a pre-2020 nonlinear macro-financial model by incorporating more recent high-volatility episodes and modern uncertainty measures. It examines whether volatility spillovers from macroeconomic uncertainty have become more asymmetric and persistent in the post-2020 environment and whether standard linear frameworks systematically understate such risks

Methodologically, the project enhances the original threshold VAR framework by integrating alternative threshold definitions, richer cross-sectional commodity data, and regime-specific impulse response analysis. The result is a modular and reproducible empirical pipeline that can be adapted for real-time macro risk monitoring by central banks, hedge funds, and international institutions

By bridging macro uncertainty, commodity volatility, and nonlinear transmission dynamics, the study delivers both an empirical update and a practical decision-support tool for financial stability analysis under elevated uncertainty.

5 Bibliography

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6 Software and Packages

- R version 4.4.3
- Packages used:
  - `tsDyn` – for threshold VAR estimation
  - `tvarGIRF` – for simulating generalized impulse response functions in TVARs
  - `svars` – for structural VAR identification
  - `vars` – for standard VAR estimation and diagnostics
  - `ggplot2` – for plotting and visualization